

ENTRY 20 — THE FIRST PREMIA PASS

Candidate `a9db39fdfab5` · **announcement_premium** · **claim_type** `premia` · **gate** `v5r3-premia` ·
PASSED, zero failures

Belt run 2026-08-23 19:26Z → 23:58Z (4h32m) · resolved by the chair 2026-08-24 · verdict row in `fund_candidates` , preserved by design

> The gate's own sentence, quoted exactly because it is precisely bounded:

> `"clears every criterion — worth a human look, which is a different claim from 'deploy it'."`

This is the first full premia-gate pass in the firm's history, and the first pass of any kind under the excess-returns machinery ($D_{23} + D_{29} + D_{32}$) that the adversary spent three kill rounds hardening. Every number below is read from the candidate's own stored verdict; nothing is recomputed here.

1 · WHAT THE CLAIM IS

Premia (constitution, 2026-08-19, amended 2026-08-21): `*better risk-adjusted return than holding the asset*`, judged over **excess returns** — both legs net of the realised risk-free series, so T-bill carry cannot impersonate edge. A premia claim does **not** need to beat buy-and-hold and must not be judged as if it should. (This one happens to anyway — see §2.) Entry 20 was ruled a premia claim by the CEO on 2026-08-23 ("Yes as premia makes sense"), with the reopening falsifier recorded: Ed's authoritative vol-ratio computation reading ≥ 1.0 reopens the ruling. It reads 0.962.

2 · THE HEADLINE NUMBERS

	STRATEGY	BENCHMARK (RECOMPUTED BASKET)
Excess Sharpe (net of realised BIL)	1.956	1.085
Raw Sharpe	2.262	1.286
Annualised volatility	14.02%	21.36%
Max drawdown	15.31%	23.88%
Total return (window)	111.45%	84.78%

CRITERION	REQUIRED	MEASURED
Sharpe advantage (excess basis)	> 0.0	+0.871
Gross exposure (engine chart, 908 obs)	≤ 1.0, fail-closed	0.9987 (max, 2025-04-24)
Drawdown not worse than the asset	true	true
Window coverage (session basis)	majority	97.9% (611 of 624 sessions)
PSR	≥ 65%	77.75%
Walkforward folds retained	≥ 50% of measurable	8 of 9 (88.9%) , median retention 0.94
Holdout retention (annualised)	≥ 0.5	1.34
Orders (priced)	≥ 20	7,001
Capacity	≥ \$100k	\$20.5M
Breakeven cost	≥ 10 bps measured	beyond the tested range (> 10 bps — a floor read; see caveat 4)

The excess conversion did its job and the candidate cleared anyway: the

raw advantage is 0.976; netting realised BIL (4.363%/yr over the window,

611 observations, fetched live — the cash leg is never snapshot-pinned)

costs the strategy's cash-holding leg ~0.105 of advantage, and +0.871

remains. The rf-breakeven is **48.85%/yr** — the premium would survive a

cash rate ten times today's before vanishing into carry.

3 · THE WALKFORWARD, FOLD BY FOLD

12 folds planned, 9 measurable, 8 retained. Three folds were **refused**

rather than fabricated — the harness declines to divide by a negative or

undersized train leg, so a loss can never report as a triumph:

FOLD	TEST END	RETENTION	NOTE
1	2022-12-29	—	refused: train leg −4.81%, no edge to retain
2	2023-04-29	—	refused: train leg −8.15%
3	2023-08-28	—	refused: train leg +3.16%, under the 5.0% measurability floor
4	2023-12-27	0.94	
5	2024-04-26	0.82	
6	2024-08-25	1.05	
7	2024-12-24	2.48	
8	2025-04-24	−0.54	the one miss — the edge inverted in the test leg
9	2025-08-23	6.27	
10	2025-12-22	0.54	
11	2026-04-22	3.74	

FOLD	TEST END	RETENTION	NOTE
------	----------	-----------	------

12	2026-08-21	0.90	
----	------------	------	--

Grid: slip $\in \{1, 3, 5, 10\}$ bps; the winner is **3 bps slip**. Required folds were scaled to the covered window (9 required for 1,818 covered days); the history floor deepened to the data path's own reach (2021-03-02, zero folds lost to it).

4 · THE GAUNTLET THIS PASS SURVIVED — WHY IT IS WORTH MORE THAN THE PASSES BEFORE IT

1. Entry 20's earlier v4.1 pass (`144387901688`) was **revoked** by gate v4.2 the same day it happened, on a breakeven read the criterion could not support. That row is preserved; a re-judge is a new row.
2. The premia **label** was killed by the adversary; the CEO re-ruled it premia with a falsifier attached (vol-ratio ≥ 1.0 reopens; reads 0.962).
3. Gate v5r1-premia was **killed** (a 4.0% rf constant where a realised series belonged). v5r2 fixed it and was **killed again** (unfinanced leverage: a zero-skill 1.25 $\times$ cash book collected free Sharpe; plus a union denominator). v5r3 closed both — exposure read from the engine's own chart, refuse-if-absent, ceiling 1.0 fail-closed — and **survived the re-blind**. That is the gate this run was judged by.
4. The run itself was submitted **snapshot-off** (`FUND_BAR_SNAPSHOT=0`) after the A/B that pinned the container-hang pathology on the snapshot layer; zero timeouts on live fetches.

5 · THE FIVE CAVEATS — FILED BESIDE THE PASS, KILL-DIRECTION ANNOTATED

1. **Survivor-only universe (direction UNKNOWN — the honest one).**

`benchmark_population` : 170 names as of 2024-02-26,
`point_in_time: false`, `survivorship_corrected: false`. Both legs draw from the same survivor universe, but the strategy also *selected* from it. The PIT membership corpus (fja05680/sp500 + the 124-name delisted Tiingo panel, 564,609 bars) exists to re-run this honestly; until then the pass is fenced to its stated population.

2. **Cash-carry understatement (CONSERVATIVE).** The engine pays 0% on idle cash while the gate subtracts realised BIL from both legs, so a cash-holding strategy leg is *understated* by $(1-\bar{w}) \cdot rf / \sigma$. The pass

happened despite this bias, not because of it. D36 is building the credit with its paired margin machinery; any re-judge creates a new row, never amends this one.

3. **PSR construction (CONSERVATIVE).** 77.75% was cleared under the current construction, whose implied target is ~1.34 Sharpe — harsher than the documented target-0 meaning. D36 restores target-0 with a calibrated level; both values will be captured going forward.

4. **Cost realism — THE ONE THAT BINDS.** Breakeven reads "beyond the tested range": a floor of >10 bps against a flat 5 bps/side assumption with 3 bps winner slip. The retro quote instrument, live since this morning, measures **real** effective spreads at 1–4 bps median on liquid names but **38–308 bps mean on small names** (SOFI, INTC). 7,001 orders across a 170-name universe must be checked against per-name measured spreads before any sizing conversation. This is the caveat that gates deployment arithmetic.

5. **Null-distance (CONSERVATIVE, stated for scale).** The adversary's zero-skill artifacts on the advantage statistic measure ~0.01, with a ±0.05 measured noise band. +0.871 sits roughly **17× outside the band** — far beyond the reach of every null measured to date.

6 · WHAT THIS IS, AND WHAT IT IS NOT

It is: the first candidate to clear every criterion of the hardened premia gate, on excess returns, inside the exposure ceiling, with an honest walkforward — the thing this firm was built to produce a truthful verdict about.

It is not: a deployment. The gate's sentence draws the line itself. The human look is the CEO's (desk item **E20-1**). If the look leads toward an experimental deployment, that path is unchanged and versioned: explicit learning goal written down, alpaca venue, exit rules committed before entry, notional capped, the CEO's click per deploy — and caveat 4 is the input to any sizing.

Filed by the CTO chair 2026-08-24. Sources: `GET /fund/factory/candidates/a9db39fdfab5` (verdict, checks, walkforward, analytics), [fund_candidates](#) row, [docs/reviews/ADVERSARY_](#) for the gate lineage, [run-cto-entry20-premia-resolve](#) for the resolution record. Per

the non-negotiables this document is never edited — a re-measurement gets
a new section or a new file.*